# CO-ORDINATE GEOMETRY


## Distance formula

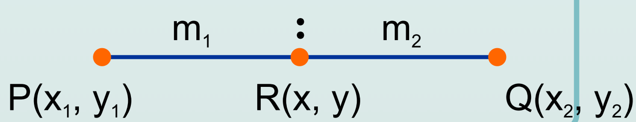

Distance between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is given by

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

## Section formula

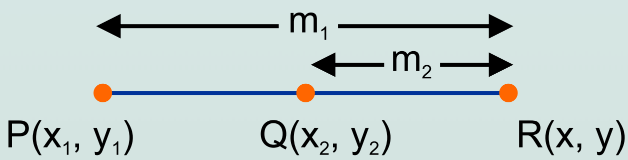
**Formula for internal division :** The coordinates of the point $R(x, y)$ which divides the join of two given points $P(x_1, y_1)$ and $Q(x_2, y_2)$ internally in the ratio $m_1 : m_2$ are given by

$$\left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$$



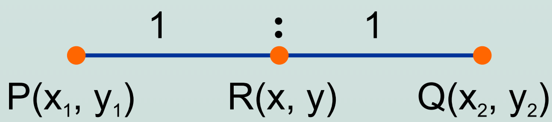

**Formula for external division :** The coordinates of the point $R(x, y)$ which divides the join of two given points $P(x_1,y_1)$ and $Q(x_2,y_2)$ externally in the ratio $m_1 : m_2$ are given by

$$\left(\frac{m_1x_2 - m_2x_1}{m_1 - m_2}, \frac{m_1y_2 - m_2y_1}{m_1 - m_2}\right)$$




**Mid-point formula :** If R is the midpoint of PQ, then its coordinates are given by

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

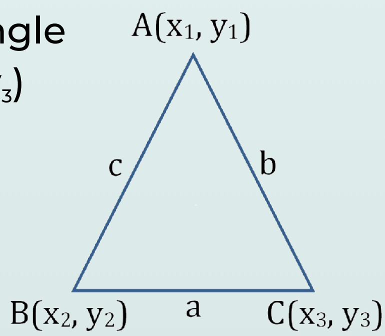


## Centroid of a triangle

The point of concurrence of the medians of a triangle is called the centroid of triangle. It divides the median in the ratio 2 : 1. The coordinates of the centroid of a triangle whose vertices are $(x_1, y_1)$, $(x_2, y_2)$ and $(x_3, y_3)$ are given by $\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$

## Incentre of a triangle

Incentre of a triangle is the point of concurrence of the internal bisectors of the angles of a triangle.
The coordinates of the incentre of a triangle whose vertices are $(x_1, y_1)$, $(x_2, y_2)$ and $(x_3, y_3)$ are given by




$$\left(\frac{ax_1+bx_2+cx_3}{a+b+c}, \frac{ay_1+by_2+cy_3}{a+b+c}\right)$$

## Area of Triangle

The area of a triangle whose vertices are $(x_1, y_1)$, $(x_2, y_2)$ and $(x_3, y_3)$ is given by

$$A = \frac{1}{2}\left|[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]\right|$$

## Condition of collinearity of three points

The points A $(x_1, y_1)$, B $(x_2, y_2)$ and C $(x_3, y_3)$ will be collinear (i.e., will lie on a sraight line) if the area of the triangle assumed to be formed by joining them is zero.